Dismantlable Digraphs and the Fixed Point Property

Rueiher Tsaur
Department of Computer Information and Network Engineering
College of Electrical Engineering and Computer Science
Lunghwa University of Science and Technology, Taoyuan County, Taiwan
rueiher@mail.lhu.edu.tw

Abstract

In this paper, the concept of continuous multifunctions for digraphs is introduced. It turns out that a digraph multifunction is continuous if and only if it is strong. We introduce the notion of dismantlable digraphs and show that it does generalize the dismantlable graphs and posets. Generalizing various previous results, we show that any dismantlable digraph has the almost fixed point property (AFPP) for strong multifunctions.

1 Power digraph and digraph multifunctions

By a directed graph (briefly *digraph*) D we mean a nonempty finite set $V(D)$ of elements called *points* and a finite set $A(G)$ of ordered pairs of distinct vertices called *arcs* (or *arrows*).. A point x of $V(D)$ is said to be *looped* if $(x,x) \in A(D)$; otherwise x is *loopless*. A digraph D is said to be *reflexive* (resp. *irreflexive*) if all its vertices are looped (resp. loopless). In this paper we concentrate solely on finite reflexive digraphs without multiple arcs. Note that it is clear that tolerance graphs and posets are digraphs.

For digraphs C and D, a *digraph homomorphism* $f : C \to D$ maps points of C to points of D, preserving the arrows, i.e., $(x,y) \in A(C) \Rightarrow (f(x), f(y)) \in A(D)$. By a *digraph multifunction* $f : C \to D$ we understand a mapping that assigns to a point $x \in V(C)$ of C a non-empty subset $f(x) \subseteq V(D)$ in D.

Definition 1.1. Let D be a digraph. The *strong power* digraph $\mathcal{P}_s(D)$ of D has the non-empty finite subsets of $V(D)$, $\mathcal{P}^+_{fin}(V(D))$ as points, and the arrow set $A_s(D)$, satisfying $(X,Y) \in A_s(D)$ if and only if $\forall x \in X, \exists y \in Y, (x,y) \in A(D)$ and $\forall y \in Y, \exists x \in X, (x,y) \in A(D)$.

Definition 1.2. The digraph multifunction f from C to D, $f : C \to D$, is *strong* if $\hat{f} : C \to \mathcal{P}_s(D)$, $\hat{f}(x) = f(x) \in \mathcal{P}^+_{fin}(V(D))$, for all $x \in V(C)$, is a digraph homomorphism.

Note that the strong digraph multifunctions reduce to digraph homomorphisms if they are single-valued. And if $f : C \to D, g : D \to E$ are strong digraph multifunctions, then the composition $g \circ f : C \to E$ will be a strong digraph multifunction. Also in [8] the power constructions for tolerance graphs and simplicial complexes were developed, and, in terms of them, the above concept of multifunctions was extended to simplicial multifunctions.

Let D be a digraph, it is clear that the strong power relation $A_s(D) = A(\mathcal{P}_s(D))$ is the conjunction of the two "weak power relations" $A_l(D), A_u(D)$ (in the notation of Brink [1]), defined by:

$$(X,Y) \in A_l(D) \Leftrightarrow \forall x \in X, \exists y \in Y, (x,y) \in A(D);$$

$$(X,Y) \in A_u(D) \Leftrightarrow \forall y \in Y, \exists x \in X, (x,y) \in A(D).$$

Note that the relations A_l and A_u correspond respectively to the so-called "lower" (or Hoare) and "upper" (or Smyth) preorders in powerdomain theory. Of course, we have $A_l(D) = (A_u(D))^{-1}$ if A is symmetric.

Definition 1.3. Let C and D be digraphs. The digraph multifunction f from C to D, $f : C \to D$, is *lower semi-strong* (resp. *upper semi-strong*) if $\hat{f} : C \to (\mathcal{P}^+_{fin}(V(D)), A_l(D))$ (resp. $\hat{f} : C \to (\mathcal{P}^+_{fin}(V(D)), A_u(D)))$ is a digraph homomorphism, where $\hat{f}(x) = f(x) \in \mathcal{P}^+_{fin}(V(D))$.

It is clear that a digraph multifunction is strong if and only if it is both lower semi-strong and upper semi-strong. Also it is easy to check that lower semi-strong and upper semi-strong digraph multifunctions are closed under compositions.

2 Continuous digraph multifunctions

Recall that a *closure space* [2] is a pair (X, cl), where $\mathrm{cl} : \mathcal{P}(X) \to \mathcal{P}(X)$ is the *closure operator* which assigns to each subset $S \subseteq X$ a subset $\mathrm{cl}(S)$ called *closure* of S, satisfies

(1) $\mathrm{cl}(\emptyset) = \emptyset$

(2) $S \subseteq \mathrm{cl}(S)$

(3) $\mathrm{cl}(S \cup T) = \mathrm{cl}(S) \cup \mathrm{cl}(T)$.

A topological space is a closure space in which the closure is idempotent, i.e., satisfying $\mathrm{cl}(\mathrm{cl}(S)) = \mathrm{cl}(S)$. On the other hand, we shall view a digraph D as a closure space by taking cl as:

$$\mathrm{cl}(S) = \{x \in V(D) \mid \exists y \in S, (x, y) \in A(D)\}.$$

The *interior operator* of a closure space (X, cl) is defined by:

$$\mathrm{int}(S) = (\mathrm{cl}(S^c))^c,$$

where c is the *complement*. Notice that in a digraph D we have:

$$\mathrm{int}(S) = \{x \in V(D) \mid B_D^+(x) \subseteq S\},$$

where $B_D^+(x) = \{y \in V(D) \mid (x, y) \in A(D)\}$.

The basic notions of topology generally extend to closure spaces, thus we have the following definition for the standard continuity conditions on many-valued functions: Let X, Y be closure spaces, and $f : X \to Y$ a multifunction. Then f is said to be *lower semi-continuous* (*lsc*) if $f(\mathrm{cl}(S)) \subseteq \mathrm{cl}(f(S))$ for all $S \subseteq X$; and *upper semi-continuous* (*usc*) if $f^+(\mathrm{int}(T)) \subseteq \mathrm{int}(f^+(T))$ for all $T \subseteq Y$, where $f^+(T) = \{x \in X \mid f(x) \subseteq T\}$.

Taking digraphs as closure spaces, we then have:

Lemma 2.1. *Let $f : C \to D$ be a digraph multifunction. Then f is lower semi-strong if and only if f is lower semi-continuous.*

Proof. Notice that, in the case of a digraph with arrow relation A, $U \subseteq \mathrm{cl}(V)$ if and only if $\forall x \in U, \exists y \in V, (x, y) \in A$, that is, if and only if $(U, V) \in A_l$. Thus, under the assumption of lower semi-continuity of f, we have: $\forall x, y \in V(C), (x, y) \in A(C) \Rightarrow (\{x\}, \{y\}) \in A_l(C) \Rightarrow \{x\} \subseteq \mathrm{cl}(\{y\}) \Rightarrow f(x) \subseteq f(\mathrm{cl}(\{y\})) \Rightarrow f(x) \subseteq \mathrm{cl}(f(y)) \Rightarrow (f(x), f(y)) \in A_l(D)$. Hence f is lower semi-strong if f is lower semi-continuous.

To show the converse, f is lower semi-strong implies: $(U, V) \in A_l(C) \Rightarrow (f(U), f(V)) \in A_l(D)$. Thus, for any $S \subseteq V(C)$, $\mathrm{cl}(S) \subseteq \mathrm{cl}(S) \Rightarrow (\mathrm{cl}(S), S) \in A_l(C) \Rightarrow (f(\mathrm{cl}(S)), f(S)) \in A_l(D) \Rightarrow f(\mathrm{cl}(S)) \subseteq \mathrm{cl}(f(S))$. Hence f is lower semi-continuous if f is lower semi-strong. ☐

Lemma 2.2. *Let $f : C \to D$ be a digraph multifunction. Then f is upper semi-strong if and only if f is upper semi-continuous.*

Proof. Let D be a digraph, and U, V subsets of $V(D)$. Then we have $(U, V) \in A_u$ if and only if $\forall y \in V, \exists x \in U, (x, y) \in A(D)$, that is, if and only if $\forall S \subseteq V(D), U \subseteq \mathrm{int}(S) \Rightarrow V \subseteq S$. Thus, f is lower semi-strong implies: $(x, y) \in A(C) \Rightarrow (f(x), f(y)) \in A_u(D) \Rightarrow [\forall T \subseteq V(D), f(x) \subseteq \mathrm{int}(T) \Rightarrow f(y) \subseteq T]$. Hence by the reflexivity of $A(C)$, we have $f^+(\mathrm{int}(T)) \subseteq \mathrm{int}(f^+(T))$.

To show the converse, suppose $(x, y) \in A(C)$ but $(f(x), f(y)) \notin A_u(D)$, that is, $\exists T \subseteq V(D)$ such that $f(x) \subseteq \mathrm{int}(T)$ but $f(y) \not\subseteq T$. Thus, $x \in f^+(\mathrm{int}(T))$ but $x \notin \mathrm{int}(f^+(T))$. Hence f is not lower semi-continuous: contradiction. ☐

Theorem 2.3. *A digraph multifunction $f : C \to D$ is strong if and only if it is continuous* (*that is, both upper and lower semi-continuous*).

Note that the following equivalence of strongness, upper semi-continuity, and lower semi-continuity is a consequence of symmetry:

Proposition 2.4 ([9, Tsaur and Smyth])**.** *Let G and H be graphs and $f : G \to H$ a multifunction. Then the following are equivalent:*

(1) *f is strong;*

(2) *f is lower semi-continuous;*

(3) *f is upper semi-continuous.*

Proof. From Lemma 2.1 and 2.2, it is clear that $(1 \Rightarrow 2)$ and $(1 \Rightarrow 3)$. To show $(2 \Rightarrow 1)$, from Lemma 2.1, f is lower semi-strong if f is lower semi-continuous. Thus by the symmetry of $E(G)$ and $E(H)$, f is upper semi-strong. The proof of $(3 \Rightarrow 1)$ is similar as above. ☐

What happens in the case of non-reflexive digraphs? In that case, the corresponding "topological" structures are not closure spaces as ordinarily understood. The same kind of analysis is still possible, however: see [7] for a discussion of "neighbourhood spaces" in relation to digraphs.

3 Dismantlable digraphs

In this section, we extend and unify several known notions of dismantlable structures such as dismantlable graphs [3] and dismantlable posets [6].

Recall that the digraph G is a (*tolerance*) *graph* if $A(G)$ is reflexive and symmetric. We will express $A(G)$ by $E(G)$ (where E stands for *edges*) if G is a graph. A finite graph G is said to be *dismantlable* if there is a linear ordering $\prec$ on $V(G)$, say $x_1 \prec x_2 \prec \cdots \prec x_m$, such that, for all $i \leq m-1$, there exists a y differs from x_i, in the subgraph G_i of G induced by $\{x_j \in V(G) \mid j \geq i\}$, satisfying $B_{G_i}(x_i) \subseteq B_{G_i}(y)$ (in this case, it is said that y *dominates* x_i in G_i, or that x_i is *dominated* by y in G_i), where $B_G(x) = \{y \mid (x,y) \in E(G)$ or $(y,x) \in E(G)\}$. Informally, the idea of dismantlability of tolerance graphs is that the graph may be contracted step by step to a point. For the continuous treatment in terms of homotopy, establishing the connection with topology, see [3, 4, 5].

The notion of dismantlable posets was first proposed in 1976 by I. Rival in his well-known paper "A fixed point theorem for finite partially ordered sets", see [6]. A *poset* is a digraph in which arrows are reflexive, anti-symmetric and transitive. We use $\leq_P$ instead of $A(P)$ if P is a poset. Following Rival [6], the point $x \in V(P)$ is said to be an *upper cover* of the point $y \in V(P)$ (or, y is a *lower cover* of x) if $y < x$ and there is no point z of P for which $y < z < x$. The point z is said to be *irreducible* of P if either z has exactly one upper cover in P or z has exactly one lower cover in P. Finally, the finite poset P is said to be $\leq$*-dismantlable* if $V(P)$ can be linearly ordered as $x_0 \prec \cdots \prec x_k$, such that for all $i \leq k-1$, x_i is irreducible in the poset P_i induced from the subset $\{x \in V(P) \mid x_i \preceq x\}$. It is clear that every $\leq$-dismantlable poset is connected (a poset is *connected* if its Hasse diagram is weakly connected).

For any point $x \in V(D)$ of the digraph D, the *outset* and the *inset* of x, denoted by $B_D^+(x)$ and $B_D^-(x)$, are defined by $B_D^+(x) = \{y \mid (x,y) \in A(D)\}$ and $B_D^-(x) = \{y \mid (y,x) \in A(D)\}$, respectively. x is said to be dominated by the point y (or y dominates x) if

(d1) $(x,y) \in A(D)$ or $(y,x) \in A(D)$,

(d2) $B_D^+(x) \backslash \{x,y\} \subseteq B_D^+(y) \backslash \{x,y\}$,

(d3) $B_D^-(x) \backslash \{x,y\} \subseteq B_D^-(y) \backslash \{x,y\}$.

As a direct generalization of the dismantlable graphs and dismantlable posets, we have:

Definition 3.1. The finite digraph D is said to be *dismantlable* if there is a linear ordering $\prec$ on $V(D)$, say $x_0 \prec \cdots \prec x_k$, such that, for all $i \leq k-1$, there exists $y, y \neq x_i$, such that y dominates x_i in the subgraph D_i of D induced by $\{x \in V(D) \mid x_i \preceq x\}$.

It is easy to check that the dismantlable graph is just a special case of the dismantlable digraph. Furthermore, by Theorem 3.2, the notion of dismantlable digraphs does generalize the dismantlable posets:

Theorem 3.2. *The poset P is dismantlable if and only if P is $\leq$-dismantlable.*

Proof. (*sufficient*) If $x \in V(P)$ is irreducible of P, that is, there exists $y \in V(P)$ such that y is the unique upper cover (or lower cover) of x. Then clearly x is dominated by y. (*necessary*) If x is dominated by y, then $(\downarrow_P x) \backslash \{x,y\} \subseteq (\downarrow_P y) \backslash \{x,y\}$ and $(\uparrow_P x) \backslash \{x,y\} \subseteq (\uparrow_P y) \backslash \{x,y\}$. Hence y is the unique upper cover (or lower cover) of x. $\square$

Recall that a digraph D is weakly connected if its underlying undirected graph is connected. Let C and D be digraphs, we denote by $\mathrm{Hom}(C,D)$ the digraph whose points are homomorphisms from C to D, with arcs $(f,g) \in A(\mathrm{Hom}(C,D))$ if and only if $(f(x), g(y)) \in A(D)$ whenever $(x,y) \in A(C)$. Finally, an induced subgraph A of the digraph D is said to be a *retract* of D if there exists a digraph homomorphism $r : D \to A$, called *retraction*, such that $r(a) = a$ for all $a \in V(A)$.

Lemma 3.3. *Let D be a finite reflexive digraph such that $\mathrm{Hom}(D,D)$ is weakly connected, and A a retract of D. Then $\mathrm{Hom}(A,A)$ is weakly connected.*

Proof. Let A be any retract of D with $r : D \to A$ the retraction. Since $\mathrm{Hom}(D,D)$ is weakly connected, each constant homomorphism x_D, is connected to id_D by a path P_x. Choose any vertex $y \in V(A)$.

It is clear that any self-mapping homomorphism $f : A \to A$ is connected to y_A by the path $r \circ P_x \circ \mathrm{im} \circ f$ (note that $f = r \circ \mathrm{id}_G \circ \mathrm{im} \circ f$ and $y_A = r \circ y_G \circ \mathrm{im} \circ f$), where $\mathrm{im} : A \to D, \mathrm{im}(x) = x$ for all $x \in V(A)$. $\square$

Theorem 3.4. *Let D be any finite reflexive digraph. Then D is dismantlable if and only if $\mathrm{Hom}(D,D)$ is weakly connected.*

Proof. ($\Rightarrow$) From the dismantlability of D, it is easy to check that there exists a path P_{x_n} from id_D to x_{nD} in $\mathrm{Hom}(D, D)$. Thus every point $f = \mathrm{id}_D \circ f$ of $\mathrm{Hom}(D, D)$ is connected to $x_{nD} = x_{nD} \circ f$. ($\Leftarrow$) It is clear that id_D is connected to every constant homomorphism x_D. Choose a minimal path, say $\{\mathrm{id}_D, f, \ldots, x_D\}$. There is a point $y \in V(D)$ satisfying $y \neq f(y) = z$. If $(y, w) \in A(D)$, then $(f(y), \mathrm{id}_D(w)) = (z, w) \in A(D)$; thus $B^+_D(y) \backslash \{y, z\} \subseteq B^+_D(z) \backslash \{y, z\}$. Similarly, $B^-_D(y) \backslash \{y, z\} \subseteq B^-_D(z) \backslash \{y, z\}$ is true. Also it is trivial that $z \in B^+_D(z)$ $(B^-_D(z))$ if $(y, z) \in A(D)$ or $(z, y) \in A(D)$. Thus y is dominated by z. It is clear that $D \backslash \{y, z\}$ is a retract of D, follow Lemma 3.3, we conclude our proof. $\square$

Corollary 3.5. *Every retract of a dismantlable digraph is dismantlable.*

4 An almost fixed point theorem for dismantlable digraphs

Given a self-mapping multifunction $f : D \to D$, a point $x \in V(D)$ is said to be a *fixed point* of f if $x \in f(x)$, and an *almost fixed point* of f if $x \in f(x)$, $(\{x\}, f(x)) \in \mathcal{P}_w(D)$, or $(f(x), \{x\}) \in \mathcal{P}_w(D)$, respectively. Let C and D be digraphs, and ϕ a property of subsets of D, then the digraph multifunction from the digraph C into the digraph D, $f : C \to D$, is said to be a ϕ-digraph multifunction if f sends points of C into subsets of D satisfying ϕ.

Definition 4.1. *Let D be a digraph. Then*

(1) *D is said to have the ϕ-*fixed point property *(ϕ-FPP) if every ϕ-digraph multifunction of D has a fixed point.*

(2) *D is said to have the ϕ-*almost fixed point property *(ϕ-AFPP) if every ϕ-digraph multifunction of D has an almost fixed point.*

Theorem 4.2. *Every dismantlable digraph has the AFPP for strong multifunctions.*

Proof. we prove by induction on $\#V(D)$ that every dismantlable digraph D has the AFPP for strong multifunctions:

Initial step. Clearly, D has the AFPP for strong multifunctions when $\#V(D) = 1$ or 2.

Inductive step. assume that D has the AFPP for strong multifunctions when $\#V(D) \leq m \in \mathcal{N}, m \geq 1$. We claim that D has the AFPP for strong multifunctions when $\#V(D) = m + 1$.

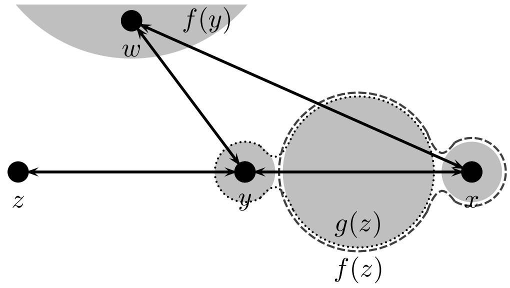

Figure 1: $x, y, z, w, g(z), f(z)$ and $f(y)$

Let $f : D \to D$ be a self-mapping strong multifunction. Since D is dismantlable, there exist $x, y \in V(D)$ such that the conditions (d1– d3) are satisfied; and the induced subgraph $D \backslash x$ is dismantlable.

We define the multifunction $g : D \backslash x \to D \backslash x$ by

$$g(u) = \begin{cases} f(u), & x \notin f(u) \\ (f(u) \backslash x) \cup \{y\}, & x \in f(u). \end{cases}$$

For any point $a \in V(D \backslash x)$, it is clear that $g(a)$ is obtained from $f(a)$ by simply replacing any occurrences of x by y. Clearly, such a replacement cannot cause strong relation $A(D \backslash x)$ of $D \backslash x$ to fail. Thus g is a strong multifunction in $D \backslash x$.

Therefore, by the inductive hypothesis, g has an almost fixed point, say $z \in V(D \backslash x)$. Then, since $f(z)$ differs from $g(z)$ by, at most, having x in place of y, therefore, z must be an almost fixed point also of f, except in the case that y is the only element of $g(z)$ satisfying $(y, z) \in A(D)$ or $(z, y) \in A(D)$, $x \in f(z)$, but $y \notin f(z)$, see Fig. 1.

We claim that y is an almost fixed point of f. Indeed, since $(f(y), f(z)) \in A(\mathcal{P}_s(D))$ or $(f(z), f(y)) \in A(\mathcal{P}_s(D))$, $f(y)$ must contain an element, say w, such that $(w, x) \in A(D)$ or $(x, w) \in A(D)$. Thus $(w, y) \in A(D)$ or $(y, w) \in A(D)$, and which implies y is an almost fixed point of f. $\square$

The following corollaries are direct consequences of Theorem 4.2:

Corollary 4.3 ([9]). *Every dismantlable graph has the AFPP for strong multifunctions.*

Corollary 4.4. *Every dismantlable ($\leq$-dismantlable) poset has the AFPP for strong multifunctions.*

Note that it is also possible to relate the AFPP for dismantlable graphs with the FPP for dismantlable posets. Recall that for any poset P, the *comparability graph* of P, denoted by comp(P), is the graph with point set $V(\text{comp}(P)) = V(P)$, such that $(x, y) \in E(\text{comp}(P))$ in comp(P) if and only if $x \leq y$ or $y \leq x$ in P. In [10], Walker showed that every $\leq$-dismantlable poset has the FPP for strong multifunctions. Thus we have

Remark 4.5. *The result of Corollary 4.3 is "stronger" than Walker's.*

Proof. Suppose the poset P is $\leq$-dismantlable. For any strong multifunction $f : P \to P$, we define the multifunction $g : \text{comp}(P) \to \text{comp}(P)$ by $g(x) = f(x), \forall x \in V(P)$. It is clear that g is a strong multifunction, and thus g has an almost fixed point, say $x \in V(P)$. Hence there exists an element $y \in f(x)$ such that $x \leq y$ or $y \leq x$, without loss of generality, we may assume the former. Now, set $x_0 = x$ and $x_1 = y$. Since f is strong, there exist $x_2 \in f(x_1)$ such that $x_1 \leq x_2$. Thus, by the same argument, we can construct a chain $\Gamma = x_0 \leq \cdots \leq x_i \leq \cdots$ in P satisfying $x_{i+1} \in f(x_i)$. Then, since P is finite, therefore it is clear that $\bigvee \Gamma$ exists and indeeed is a fixed point of f. □

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